SPRiF: Spectral Projective Reset Integrate-and-Fire Neuron

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Abstract

Long-term temporal processing requires spiking neural networks (SNNs) to preserve information across delays while continuing to fire. Standard spiking neurons use one membrane variable to integrate history, generate spikes, and receive post-spike reset, so communication can perturb retained context. We introduce **SPRiF** (Spectral Projective Reset Integrate-and-Fire Neuron), which separates these functions. A constrained spectral slow state represents context through real and damped rotational modes; a fast state generates spikes and receives a learnable projective reset. The slow state has no direct local reset update, although emitted spikes still propagate through recurrent connections. Across five temporal classification benchmarks, SPRiF records the highest reported mean accuracy among the listed methods on three tasks and remains competitive on the other two; because the published systems use task-specific protocols, these results are descriptive comparisons. We also introduce Spike-Intervention Delayed Match-to-Sample (SI-DMS), which measures paired recall loss when additional spikes trigger native reset and recurrent propagation. In an unseen setting with 40 intervention events applied to 15% of hidden units per event, SPRiF loses 0.2 ± 0.2 percentage points relative to its clean accuracy. Impulse-response analysis identifies heterogeneous learned temporal kernels. Functional state decomposition therefore provides a practical neuron-level design for the tested long-term temporal tasks.

Introduction

Spiking neural networks (SNNs) encode and transmit temporal information through discrete spike events. Their neuron dynamics must preserve relevant context while continuing to generate those events, a requirement that becomes difficult when dependencies span multiple timescales.

The leaky integrate-and-fire (LIF) neuron remains widely used because it is simple and compatible with surrogate-gradient training. Its extensions introduce adaptive variables or thresholds (Baronig et al. 2025; Zhang et al. 2025b), resonant dynamics (Higuchi et al. 2024; Huang et al. 2024; Zhang et al. 2025a), heterogeneous temporal decays (Wei et al. 2026; Deng et al. 2025; Fulleda-Garcia, Soldado-Magraner, and Margarit-Taulé 2026; Tiwari and Chauhan 2025), and modified reset (Guo et al. 2022; Huang et al. 2026; Li et al. 2025; Fang et al. 2023). These mechanisms enrich the available temporal responses, but the spike-generating variable usually remains both the memory carrier and the reset target.

Reset therefore alters the same recurrence that transports input history. This functional coupling, rather than limited timescale diversity alone, motivates separating temporal memory from resettable discharge without sacrificing binary communication or enlarging the network architecture.

We introduce **SPRiF** (Spectral Projective Reset Integrate-and-Fire Neuron), which implements this separation through two state groups. A slow spectral state combines one real exponential mode with a two-dimensional damped rotational mode to store temporal context. A fast discharge state projects this context to a membrane readout, emits binary spikes, and receives a learnable projective reset along $[1, \lambda]^\top$. SPRiF is trained with backpropagation through time and surrogate gradients; reset acts directly on the fast state while the slow state continues its spectral evolution.

SPRiF differs from multi-timescale models that add parallel decay channels without separating state functions. Its slow state stores temporal context, while its fast state handles spike readout and reset. The defining distinction is the absence of a direct reset path to the slow state, not simply the presence of heterogeneous time constants.

We evaluate SPRiF on five benchmarks spanning sequential images, physiological signals, speech, and event-based digits. Among the listed published results, SPRiF has the highest reported mean accuracy on S-MNIST, PS-MNIST, and QTDB and remains competitive on GSC and SHD. Its reported end-to-end parameter count is lower than that of the highest-accuracy listed baseline on four datasets and comparable on QTDB. Mechanism ablations, recurrent spike intervention, and impulse-response analyses examine the roles of rotation, state separation, and projective reset.

The main contributions of our work are summarized below.

1. We formulate functional state decomposition as a neuron design in which temporal memory and spike-triggered reset occupy distinct states.
2. SPRiF instantiates this principle with a constrained spectral slow state that combines real exponential decay and

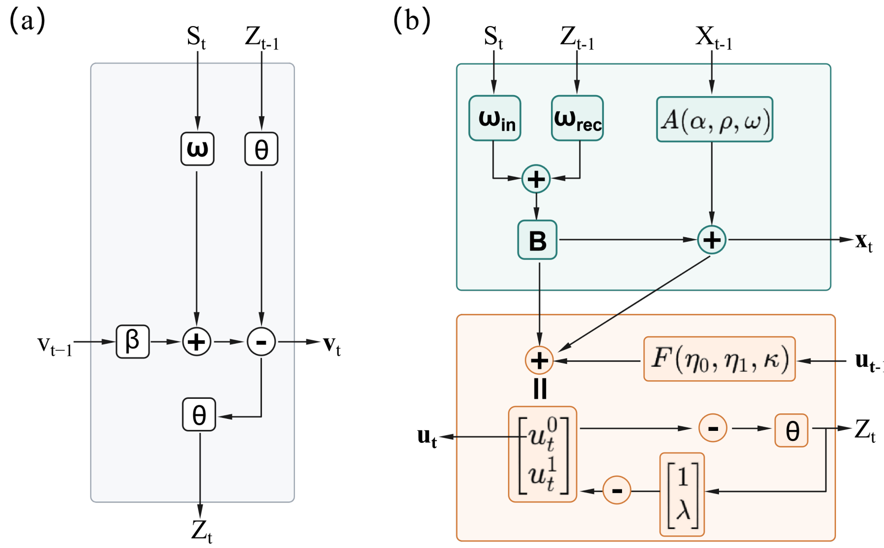

Figure 1: Architectures of (a) LIF and (b) SPRiF. SPRiF separates a slow memory state with no direct local reset update from a resettable fast discharge state.

damped rotational modes, together with a fast discharge state that receives a learned projective reset.

3. We evaluate SPRiF on five temporal benchmarks and probe its mechanism with three ablations, SI-DMS, and analyses of learned temporal kernels. Code will be released upon acceptance.

Related Work

Neuron Dynamics for Temporal Processing

Adaptive variables, heterogeneous decays, and temporal gates broaden the response repertoire of LIF neurons (Yin, Corradi, and Bohté 2021; Baronig et al. 2025; Deng et al. 2025; Tiwari and Chauhan 2025). These approaches enrich temporal dynamics without imposing SPRiF's neuron-local division between spectral memory and resettable discharge.

Resonate-and-fire variants encode damped oscillations (Izhikevich 2001; Higuchi et al. 2024; Huang et al. 2024; Zhang et al. 2025a), while heterogeneous neurons and synapses distribute temporal decays across a network (Perez-Nieves et al. 2021; Deng et al. 2025). Compartmental models add dendritic structure and dynamics (Zhang et al. 2024; Zheng et al. 2024; Pagkalos, Chavlis, and Poirazi 2023); LSTM-LIF, for example, separates a dendritic long-term memory compartment from somatic discharge (Zhang et al. 2023). Structured state-space dynamics provide another route to long-horizon processing (Fabre et al. 2025; Shen et al. 2025). DMP-SNN combines layer-shared, low-dimensional memory with a fast feed-forward spiking pathway (Sun et al. 2026). SPRiF instead places spectral memory and projective reset in distinct state groups within each neuron.

Reset and Functional State Separation

Existing mechanisms soften, adapt, or remove the post-spike update, and recent work also revisits reset and refractory periods through binary-activated recurrent sequence models (Guo et al. 2022; Li et al. 2025; Huang et al. 2026; Fang et al. 2023; Zhang 2025). These designs modify the spike-generating state or remove its reset. SPRiF retains reset but confines it to the fast discharge state, leaving the slow spectral state outside the local reset path. Network-level temporal processing and robustness strategies (Ding et al. 2025b, 2024, 2025a) remain complementary to this neuron-level design.

Method

Limitation of Conventional LIF Dynamics

For presynaptic activities $s_{i,t}$, a standard leaky integrate-and-fire (LIF) neuron receives input current

$$I_t = \sum_i w_i s_{i,t} + b, \tag{1}$$

where w_i is the synaptic weight of input i and b is a bias term. Its membrane update is

$$v_t = \beta v_{t-1} + I_t - z_{t-1}\theta, \tag{2}$$
$$z_t = H(v_t - \theta). \tag{3}$$

Here, $\beta \in (0, 1)$ is the leak factor, H is the Heaviside step function, and θ is the firing threshold. The scalar v_t carries temporal context, determines spike timing, and receives the preceding reset. Unrolling over k steps gives

$$v_t = \beta^k v_{t-k} + \sum_{\ell=0}^{k-1} \beta^\ell \big(I_{t-\ell} - \theta z_{t-\ell-1}\big). \tag{4}$$

Input and reset histories are filtered through the same recurrence, so reset is inseparable from the state that transports temporal information. This motivates distinct memory and post-spike discharge states.

SPRiF Spiking Neuron Model

Figure 1 contrasts the two architectures. For layer input $\mathbf{s}_t$, the input current is

$$I_t = W_{\text{in}}\mathbf{s}_t + W_{\text{rec}}\mathbf{z}_{t-1} + b, \tag{5}$$

where the recurrent term is omitted in feed-forward layers. In the single-neuron equations below, I_t denotes the scalar current received by that neuron. Each neuron j maintains a three-dimensional slow state $\mathbf{x}_t \in \mathbb{R}^3$ and a two-dimensional fast state $\mathbf{u}_t \in \mathbb{R}^2$; $\widetilde{\mathbf{u}}_t$ denotes the pre-reset fast state. Its dynamics are

$$\begin{cases} \mathbf{x}_t = A\mathbf{x}_{t-1} + BI_t, \\ \widetilde{u}_t^0 = \eta_0 u_{t-1}^0 + \kappa u_{t-1}^1 + (G\mathbf{x}_t)^0, \\ \widetilde{u}_t^1 = \eta_1 u_{t-1}^1 + (G\mathbf{x}_t)^1, \\ v_t = \widetilde{u}_t^0, \qquad z_t = H(v_t - \theta), \\ \mathbf{u}_t = \widetilde{\mathbf{u}}_t - z_t\theta \begin{bmatrix} 1 \\ \lambda_j \end{bmatrix}. \end{cases} \tag{6}$$

The update follows $I_t \to \mathbf{x}_t \to \widetilde{\mathbf{u}}_t \to v_t \to z_t \to \mathbf{u}_t$. The slow state is updated before spike generation; the fast state is then read out, thresholded, and reset. The direct local reset update therefore acts only on the fast path, although $W_{\text{rec}}\mathbf{z}_t$ lets spikes influence the slow state at later timesteps in recurrent layers.

Equation (6) is written for one neuron, with indices suppressed except in λ_j. Except for W_{in} and W_{rec}, the listed dynamics parameters are per-neuron. Layers apply the update at each timestep, and recurrence is optional.

Spectral Slow-Memory Dynamics

The slow-state update in Eq. (6) uses a three-dimensional real-valued system with one real decay mode and one complex-conjugate rotational pair. This constrained transition retains temporal context while keeping its spectrum interpretable.

$$A = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & \rho\cos\omega & -\rho\sin\omega \\ 0 & \rho\sin\omega & \rho\cos\omega \end{bmatrix}, \quad B = \begin{bmatrix} 1-\alpha \\ 1-\rho \\ 0 \end{bmatrix}. \tag{7}$$

The spectrum of A is

$$\operatorname{eig}(A) = \left\{\alpha, \rho e^{\mathrm{i}\omega}, \rho e^{-\mathrm{i}\omega}\right\}. \tag{8}$$

The first coordinate is an exponential trace, $x_t^0 = \alpha x_{t-1}^0 + (1-\alpha)I_t$, while (x_t^1, x_t^2) form a damped rotation driven through the first oscillatory coordinate. We parameterize

$$\alpha = \sigma(\alpha_{\text{raw}}), \quad \rho = \sigma(\rho_{\text{raw}}), \quad \omega = \pi\sigma(\omega_{\text{raw}}), \tag{9}$$

where σ is the logistic sigmoid. Thus $\alpha, \rho \in (0,1)$ and $\omega \in (0,\pi)$ place every eigenvalue of A strictly inside the unit disk. The parameters α, ρ, and ω control real decay, rotational damping, and rotational frequency, respectively, yielding a stable, interpretable spectrum.

The resulting temporal basis is explicit. For a unit impulse entering the slow state, the response after k subsequent transitions is

$$A^k B = \begin{bmatrix} (1-\alpha)\alpha^k \\ (1-\rho)\rho^k\cos(k\omega) \\ (1-\rho)\rho^k\sin(k\omega) \end{bmatrix}. \tag{10}$$

The real coordinate supplies a monotone exponential trace, whereas the rotational pair supplies phase-shifted, damped oscillations. They form a compact temporal basis with distinct decay and frequency controls.

Fast Discharge Dynamics and Projective Reset

The fast-state equations in Eq. (6) read out slow context and implement resettable discharge. The learned coefficients $\eta_0, \eta_1 \in (0,1)$ are fast leaks, κ couples the second fast coordinate to the first, and $G \in \mathbb{R}^{2\times 3}$ projects slow dynamics into discharge space.

The first fast coordinate is the membrane potential $v_t = \widetilde{u}_t^0$, and the forward pass generates $z_t = H(v_t - \theta)$ with a hard threshold. During optimization, only the threshold derivative is replaced by the multi-Gaussian surrogate (Yin, Corradi, and Bohté 2021); the forward dynamics remain unchanged.

After a spike, SPRiF applies the following projective reset only to the fast state.

$$\mathbf{u}_t = \widetilde{\mathbf{u}}_t - z_t\theta \begin{bmatrix} 1 \\ \lambda_j \end{bmatrix}, \tag{11}$$

where λ_j is learned independently for neuron j. Setting $\lambda_j = 0$ recovers scalar subtraction on the membrane coordinate; otherwise, reset modifies both fast coordinates along a learned direction. Thus, λ_j controls post-spike correction without changing the slow-state transition.

Reset-Memory Separation Analysis

The preceding equations separate the same-step effect of reset from the later propagation of spike signals. Relative to the pre-reset states, the reset-induced increments are

$$\Delta\mathbf{x}_t^{\text{reset}} = \mathbf{0}, \qquad \Delta\mathbf{u}_t^{\text{reset}} = -z_t\theta \begin{bmatrix} 1 \\ \lambda_j \end{bmatrix}. \tag{12}$$

In LIF, the reset contribution $-z_{t-1}\theta$ is accumulated through the same recurrence as input history. In SPRiF, Eq. (12) gives a zero same-step increment for $\mathbf{x}_t$ and confines reset to $\mathbf{u}_t$.

The slow-state recurrence makes this distinction explicit. Iterating the first line of Eq. (6) yields

$$\mathbf{x}_t = A^t\mathbf{x}_0 + \sum_{\tau=1}^{t} A^{t-\tau}BI_\tau. \tag{13}$$

Holding I_t fixed, Eq. (13) gives $\partial\mathbf{x}_t/\partial z_t = \mathbf{0}$ for the reset operation. In recurrent layers, $\mathbf{z}_t$ can still affect $\mathbf{x}_{t+1}$ through $W_{\text{rec}}\mathbf{z}_t$. The construction therefore distinguishes same-step local reset from later network communication.

Experiments

Benchmarks and Protocol

We evaluate SPRiF on five temporal classification benchmarks spanning sequential images (S-MNIST and PS-MNIST), physiological signals (QTDB), frame-based speech (GSC), and event-driven speech (SHD). We train all models for a fixed number of epochs. The tasks differ in input encoding, sequence length, and temporal structure and therefore test the neuron dynamics across several temporal modalities.

Table 1: Comparison across five temporal benchmarks. "Rec" denotes recurrent synaptic connections; DMP-SNN uses layer-shared state-space memory. Parameters are reported in millions (M), and SPRiF entries (bold) are mean±standard deviation over five seeds.

Datasets	Method	Network	Parameters (M)	Accuracy (%)
S-MNIST	GLIF$_{\text{NeurIPS, 2022}}$ (Yao et al. 2022)	Rec	0.15	96.64
	ASRNN$_{\text{Nat. Mach. Intell., 2021}}$ (Yin, Corradi, and Bohté 2021)	Rec	0.15	98.70
	DH-SNN$_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024)	Rec	0.08	98.90
	BRF$_{\text{ICML, 2024}}$ (Higuchi et al. 2024)	Rec	0.069	99.10
	TC-LIF$_{\text{AAAI, 2024}}$ (Zhang et al. 2024)	Rec	0.15	99.20
	DMP-SNN (II)$_{\text{Nat. Mach. Intell., 2026}}$ (Sun et al. 2026)	DMP	0.073	99.20
	SPRiF (ours)	**Rec**	**0.067**	**99.28 ± 0.06**
PS-MNIST	GLIF$_{\text{NeurIPS, 2022}}$ (Yao et al. 2022)	Rec	0.15	90.47
	ASRNN$_{\text{Nat. Mach. Intell., 2021}}$ (Yin, Corradi, and Bohté 2021)	Rec	0.15	94.30
	DH-SNN$_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024)	Rec	0.08	94.52
	BRF$_{\text{ICML, 2024}}$ (Higuchi et al. 2024)	Rec	0.069	95.20
	TC-LIF$_{\text{AAAI, 2024}}$ (Zhang et al. 2024)	Rec	0.15	95.36
	DMP-SNN (I)$_{\text{Nat. Mach. Intell., 2026}}$ (Sun et al. 2026)	DMP	0.061	95.50
	SPRiF (ours)	**Rec**	**0.067**	**95.86 ± 0.14**
QTDB	ASRNN$_{\text{Nat. Mach. Intell., 2021}}$ (Yin, Corradi, and Bohté 2021)	Rec	0.0018	85.90
	DH-SNN$_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024)	Rec	0.00178	86.35
	BRF$_{\text{ICML, 2024}}$ (Higuchi et al. 2024)	Rec	0.00173	87.00
	SE-adLIF$_{\text{Nat. Commun., 2025}}$ (Baronig et al. 2025)	Rec	0.0018	86.88
	SPRiF (ours)	**Rec**	**0.00177**	**88.43 ± 0.36**
GSC	SNN-SFA$_{\text{eLife, 2021}}$ (Salaj et al. 2021)	Rec	4.19	91.21
	ASRNN$_{\text{Nat. Mach. Intell., 2021}}$ (Yin, Corradi, and Bohté 2021)	Rec	0.31	92.10
	RadLIF$_{\text{Front. Neurosci., 2022}}$ (Bittar and Garner 2022)	Rec	1.20	94.51
	DH-SNN$_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024)	Rec	0.13	93.80
	TC-LIF$_{\text{AAAI, 2024}}$ (Zhang et al. 2024)	Rec	0.20	94.84
	SPRiF (ours)	**Rec**	**0.13**	**94.55 ± 0.31**
SHD	Heterogeneous SNN$_{\text{Nat. Commun., 2021}}$ (Perez-Nieves et al. 2021)	Rec	0.11	82.50
	ASRNN$_{\text{Nat. Mach. Intell., 2021}}$ (Yin, Corradi, and Bohté 2021)	Rec	0.14	90.40
	DH-SNN$_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024)	Rec	0.05	91.34
	BRF$_{\text{ICML, 2024}}$ (Higuchi et al. 2024)	Rec	0.11	91.70
	TC-LIF$_{\text{AAAI, 2024}}$ (Zhang et al. 2024)	Rec	0.14	88.91
	MPS-SNN$_{\text{ICLR, 2025}}$ (Ding et al. 2025b)	Rec	0.39	91.19
	DGN$_{\text{ICLR, 2026}}$ (Bai, Wang, and Yu 2026)	Rec	0.22	87.78
	SPRiF (ours)	**Rec**	**0.05**	**91.52 ± 0.46**

We compare SPRiF with published recurrent SNNs that use gated, adaptive, dendritic, resonate-and-fire, and other temporal neuron dynamics. The architectures and optimization protocols are adapted to each benchmark, whereas the reported SPRiF variants retain the same neuron definition. Complete dataset preprocessing, task-specific architectures, optimization settings, initialization procedures, and reproducibility details are provided in the technical appendix. Classification accuracy is reported in Table 1.

Main Results

Table 1 summarizes the benchmark comparison. SPRiF obtains the highest reported accuracy among the listed methods on S-MNIST, PS-MNIST, and QTDB and remains competitive on GSC and SHD. Its reported parameter count is lower than that of the highest-accuracy listed baseline on four datasets and comparable on QTDB. Because the published systems use task-specific architectures and training protocols, these values provide descriptive context rather than a matched significance or efficiency comparison.

Mechanism Ablations

We isolated three design choices on PS-MNIST, GSC, and QTDB. Ablation A set $\omega = 0$, removing rotational coupling while retaining the real decay path. Ablation B merged memory and discharge into a single three-dimensional state, whose first coordinate served as both membrane and reset target. Ablation C fixed $\lambda = 0$, reducing Eq. (11) to scalar reset. All variants otherwise retained the task-specific training configuration.

Table 2 shows the same ordering on all three datasets. The merged variant has the largest observed accuracy decrease,

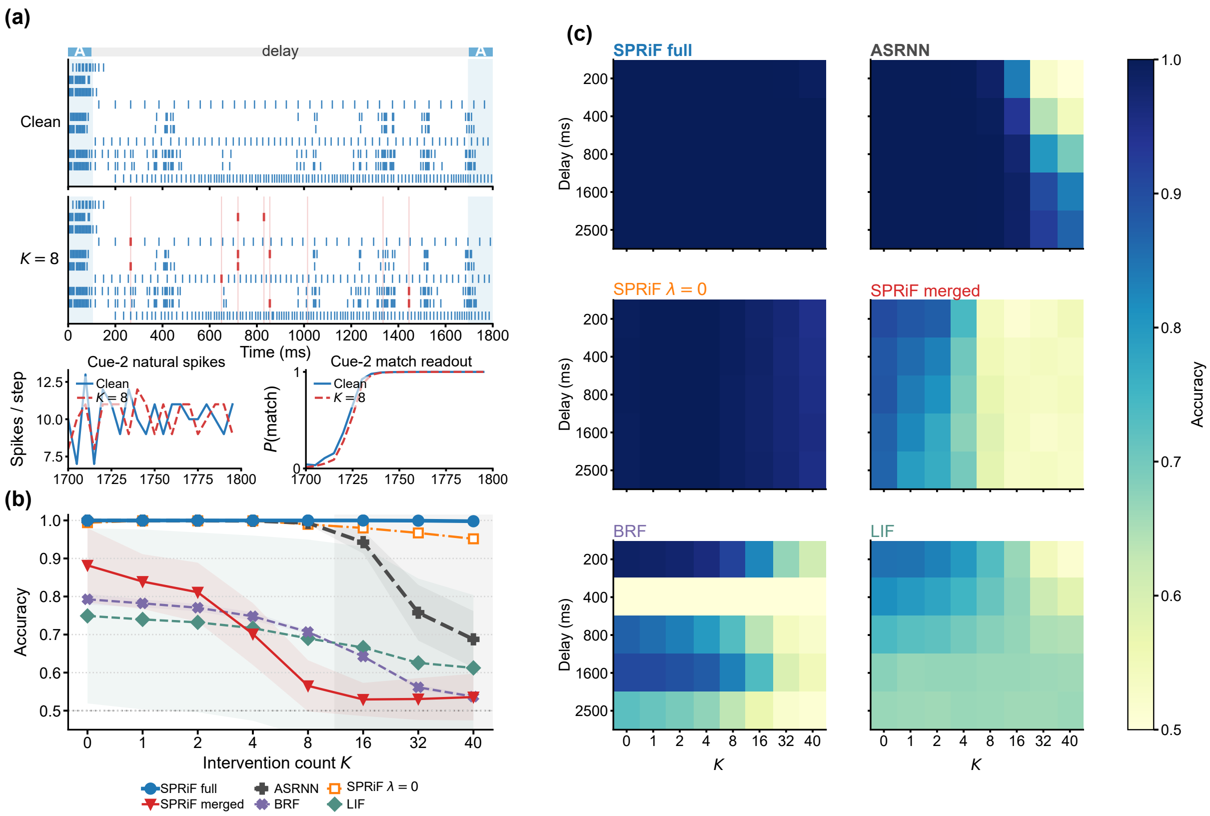

Figure 2: SI-DMS at the fixed 15% intervention fraction. (a) Paired clean and $K = 8$ replays of the same deterministic A–A input for full SPRiF at a 1600-ms delay. The cues share stimulus identity but use independent spike realizations. Blue and red ticks mark natural and forced spikes; faint red lines mark the eight intervention times. Cue-2 details show natural activity and the match readout; both conditions predict match. (b) Accuracy versus K, averaged over five delays within each of three seeds; bands show seed standard deviations. (c) Three-seed mean delay-by-K accuracy with a shared color scale. Training includes $K \leq 8$; $K \in \{16, 32, 40\}$ are unseen.

Table 2: Mechanism ablations (accuracy, %).

Dataset	Full*	ω=0	Merged	λ=0
PS-MNIST	**95.86**	92.29	93.44	95.32
GSC	**94.55**	93.83	90.05	94.22
QTDB	**88.43**	87.78	83.39	87.40

*Full values are the five-seed mean from Table 1.

2.42–5.04 percentage points. The ω=0 variant is 0.65–3.57 percentage points below full SPRiF, with the largest difference on the long permuted sequence, and the λ=0 variant is 0.33–1.03 percentage points lower. Because merging also reduces total state dimensionality, its larger decrease cannot be attributed to state separation alone. The controlled ω=0 and λ=0 variants yield smaller differences under the tested settings.

Controlled Spike Intervention

SI-DMS separates task acquisition from vulnerability to additional spike-triggered updates. It reports clean accuracy together with paired recall loss after forced spikes trigger native reset and recurrent propagation during the memory delay.

The test uses a 64-unit recurrent hidden layer. During the delay, 15% of hidden neurons are induced to spike at each of K sampled timesteps, followed by a match/non-match probe. Placing the membrane coordinate above threshold before spike computation triggers native reset, and the emitted spike enters the next-step recurrent pathway. SI-DMS therefore measures combined spike-reset and recurrent-spike stress, not a pure local-reset effect. Training samples $K \in \{0, 1, 2, 4, 8\}$; evaluation adds $K \in \{16, 32, 40\}$ and sweeps the intervention fraction from 5% to 50%.

Figure 2a replays the same deterministic A–A input under clean and $K = 8$ conditions. After the 1600-ms delay and eight forced events, the stressed replay retains natural cue-2 activity and the same correct match decision. This ex-

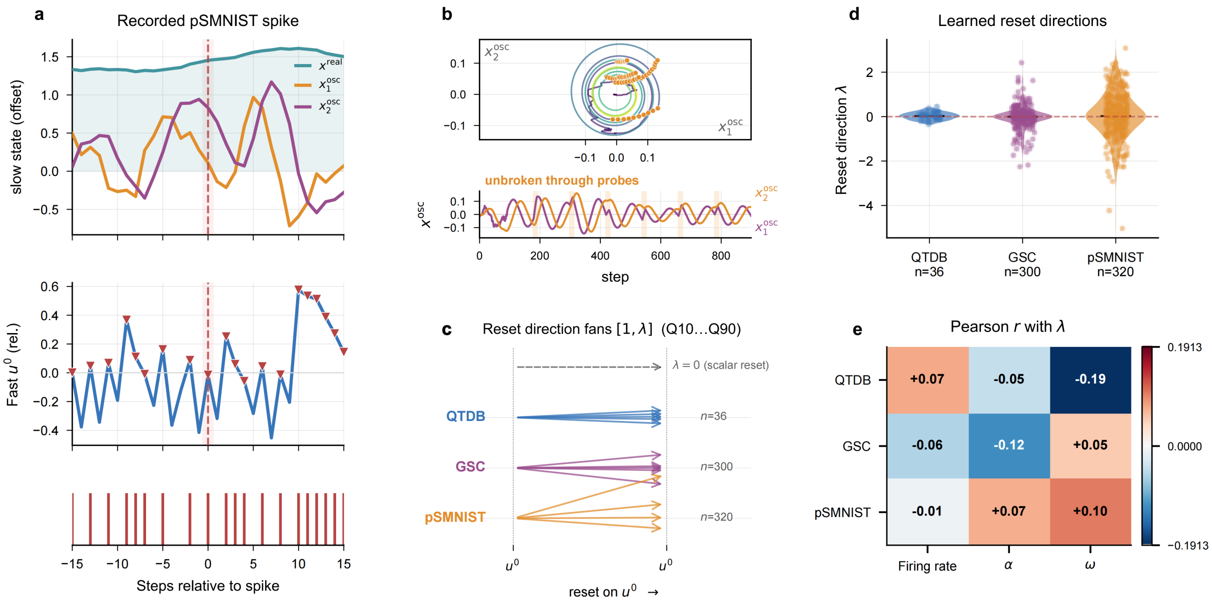

Figure 3: Functional decomposition and learned projective reset. (a) Slow and fast states around a recorded PS-MNIST spike. (b) Oscillatory slow-state trajectories across cue–delay probe windows. (c) Representative reset directions $[1, \lambda]$. (d) Task-wise distributions of λ. (e) Pearson correlations between λ and firing rate, α, and ω.

ample illustrates the process; Fig. 2b–c provides population evidence across models, delays, and seeds.

For each seed and delay, $\Delta\mathrm{Acc}(K, f) = \mathrm{Acc}(0) - \mathrm{Acc}(K, f)$. Table 3 averages over five delays within each seed and then over three seeds at $f = 15\%$. ASRNN provides the clean-matched control: full SPRiF and ASRNN both attain 100.0±0.0 clean accuracy, but their losses at unseen $K = 40$ are 0.2±0.2 and 31.2±7.4 percentage points, respectively. BRF and LIF start lower, with LIF also showing substantial seed variability, so clean accuracy and paired loss should be read together.

At $K = 40$, the λ=0 variant loses 4.2±0.2 percentage points, compared with 0.2±0.2 for full SPRiF, supporting a task-specific benefit from the learned reset direction. Both separated variants are more resilient than the merged model, whose losses are 32.3±1.4 at $K = 8$ and 36.4±2.7 at $K = 40$, alongside lower clean accuracy. Because merging also reduces state dimensionality, this result cannot isolate separation from capacity. Figure 2 gives the complete delay-by-K view; the appendix provides the fraction sweep and integrity audit.

Dynamical Analysis

The update equations give $\mathbf{x}_t$ a zero same-step reset increment, while recurrent SI-DMS measures recall after forced spikes undergo native reset and enter the next-step recurrent pathway. Figure 3a records slow and fast states around a PS-MNIST spike, and Fig. 3b follows slow-state oscillation across cue–delay probe windows. These traces visualize continuous slow-state evolution; SI-DMS supplies the intervention-based evidence. Figure 3c–e characterizes the learned projective reset directions.

Table 3: SI-DMS clean accuracy and paired intervention loss at $f = 15\%$ (mean±standard deviation over three seeds; percentage points). Each seed-level value is averaged over five delays before aggregation.

Model	Clean	$\Delta\mathrm{Acc}_8$	$\Delta\mathrm{Acc}_{40}$
SPRiF full	100.0±0.0	**0.0±0.0**	**0.2±0.2**
SPRiF λ=0	99.4±0.0	0.4±0.1	4.2±0.2
SPRiF merged	88.1±1.4	32.3±1.4	36.4±2.7
ASRNN	100.0±0.0	0.6±0.5	31.2±7.4
BRF	79.3±1.4	8.6±1.4	25.6±1.2
LIF	74.9±23.0	5.9±8.2	13.7±10.2

The learned reset directions do not collapse to a common value. On QTDB, GSC, and PS-MNIST, λ spans both signs (Fig. 3c–d), and its correlations with firing rate, α, and ω remain weak, with $|r| \leq 0.191$ (Fig. 3e). Combined with the λ=0 ablation, this variation shows that training learns heterogeneous reset directions in the fast state. The result characterizes reset geometry in the tested networks and does not imply a biological interpretation.

The impulse-response analysis complements SI-DMS by exposing the intrinsic slow transition rather than recall under induced spikes. We inject a unit impulse into trained neurons and record 100 steps of slow-state evolution without subsequent input, isolating the temporal basis $A^k B$ in Eq. (10). Figure 4 combines trajectories sampled across α quantiles with real-coordinate spectra aggregated over all neurons.

The α-stratified real coordinates span rapid and persistent

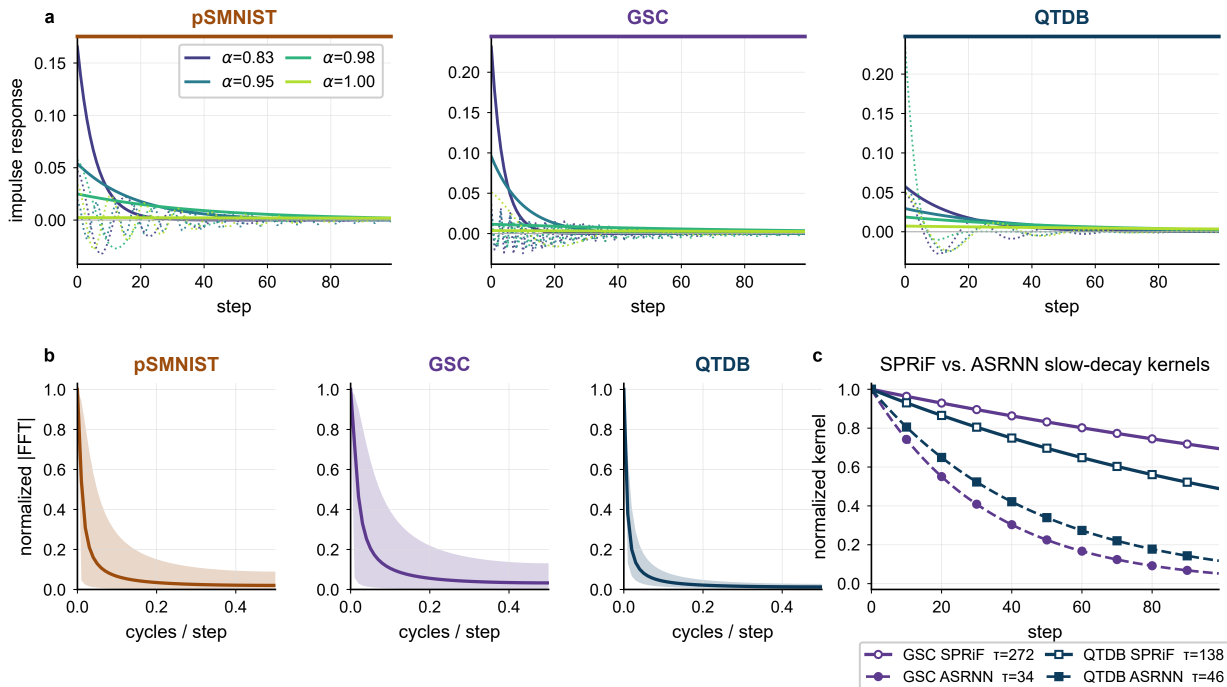

Figure 4: Learned temporal kernels. (a) Real (solid) and oscillatory (dotted) coordinates of four α-quantile SPRiF neurons per task over 100 post-impulse timesteps. (b) Median normalized real-coordinate spectrum with the 10–90% range. (c) Representative SPRiF real-coordinate kernels and membrane kernels from the ASRNN checkpoints available for GSC and QTDB, with derived time constants τ.

responses, and the rotational coordinates vary in phase and frequency. At the population level, the median normalized real-coordinate spectra concentrate at low frequencies, with substantial within-task variation in the 10–90% ranges. Normalization removes absolute amplitude from this comparison. The resulting response families reflect the constrained parameterization: α controls monotone retention, and ρ and ω govern rotational damping and phase evolution.

Available checkpoints permit the cross-model kernel comparison only on GSC and QTDB. On these tasks, selected slow-decay SPRiF real modes have derived time constants of approximately 272 and 138 steps, respectively, compared with 34 and 46 steps for the trained ASRNN membrane kernels. We treat this as a two-task diagnostic, not a general cross-model advantage. The appendix provides additional impulse-response and spectral analyses.

Conclusion

SPRiF separates temporal memory from spike-triggered discharge: a constrained spectral slow state carries real and rotational modes, and a fast state emits binary spikes and receives projective reset. Across five temporal benchmarks, SPRiF has the highest reported mean accuracy among the listed methods on three tasks and remains competitive on the other two; because the published systems use different protocols, these cross-paper results provide descriptive context. At the unseen $K = 40$ setting in recurrent SI-DMS, full SPRiF loses $0.2{\pm}0.2$ percentage points, compared with $31.2{\pm}7.4$ for clean-matched ASRNN and $4.2{\pm}0.2$ for the $\lambda{=}0$ variant. The merged variant shows the largest benchmark degradation but also reduces state dimensionality, so it does not isolate separation from capacity. Under the tested conditions, functional state decomposition provides a practical way to protect temporal context, with the learned reset direction offering task-specific refinement; evaluation at larger scale and on neuromorphic hardware remains future work.

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